

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course:B.Tech

Branch : Artificial Intelligence and Data Science

Semester : III

Subject Code & Name: BTAIES305_Y23 Digital Logic & Signal Processing

Max Marks: 60

Date:21/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The octal equivalent of the decimal number $(417)_{10}$ is ____	BT 1/CO1	1
	a. $(641)_8$ b. $(642)_8$ c. $(461)_8$ d. $(462)_8$		
2	Perform binary addition: $101101 + 011011 = ?$	BT2/CO1	1
	a. 011010 b. 1010100 c. 101110 d. 1001000		
3	The NOR gate output will be high if the two inputs are ____	BT1/CO1	1
	a. 0 0 b. 0 1 c. 1 0 d. 1 1		
4	Discrete time signal is derived from continuous time signal by ____ process.	BT1/ CO2	1
	a. Addition b. Multiplying c. Sampling d. Encoding		
5	Energy signal has __ average power and power signal has __ energy.	BT2/ CO2	1
	a. 0,0 b. ∞, ∞ c. 0, ∞ d. $\infty, 0$		
6	A signal is a physical quantity which does not vary with ____	BT1/ CO2	1
	a. Time b. Space c. Dependent Variable d. Independent Variable		
7	If $x(n)$ and $X(k)$ are an N-point DFT pair, then $X(k+N)=?$	BT1/ CO3	1
	a. $X(-k)$ b. $-X(k)$ c. $X(k)$ d. None of the mentioned		
8	D.I.T. is	BT2/ CO3	1
	a.Degradation In Time b. Decimation In Time c.Disadvantage In Time d. All of the above		
9	Which of the following method is used to find the inverse z-transform of a signal?	BT1/ CO4	1

	a. Long Division Method	b. Partial fraction Expansion	c. Residue Method	d. All of the above		
10	What is the set of all values of z for which X(z) attains a finite value?				BT2/ CO4	1
	a. Region of convergence	b. Ration of Conversion	c. Infinite Conversion	d. None of them		
11	Which function has a provision of determining the similarity between the signal and its delayed version?				BT1/ CO5	1
	a. Auto-correlation	b. Cross-correlation	c. Convolution	d. DFT		
12	Power spectral density function is a?				BT2/ CO5	1
	a. Real and even function	b. Non negative function	c. Periodic	d. All of the mentioned		
Q. 2	Solve the following.					12
A)	Construct (2048) ₁₀ in Straight binary, Natural BCD, Excess-3, Gray, Octal and Hexadecimal codes.				BT3/CO1	6
B)	Classify & explain Logic Gates with symbol and truth table.				BT1/CO1	6
Q.3	Solve the following.					12
A)	Define, classify and explain with examples "Systems".				BT2/CO2	6
B)	Sketch following signals and perform operations $x(n) = \begin{cases} 1 & \text{for } -2 < n < 3 \\ 2 & \text{for } n = 4 \text{ to } 5 \\ 0 & \text{for otherwise} \end{cases}$ i. x(n-2) + x(n+1) ii. -x(-n-2) + x(-n+1) iii. Even & Odd component				BT4/CO2	6
Q. 4	Solve Any Two of the following.					12
A)	Compute 4-point DFT of x(n) = {1,1,2,2}				BT3/CO3	6
B)	State and prove any three properties of DFT.				BT1/CO3	6
C)	Perform circular convolution of x(n)={1,1,2,2,3,3} and h(n)={1,1,2,2,2}				BT3/CO3	6
Q.5	Solve Any Two of the following.					12
A)	Compute Z- Transform of x(n) = {1,2,1,0,1,2,1}				BT2/CO4	6
B)	Find IZT using Partial Fraction Expansion Method. $X(z) = \frac{1+3z^{-1}}{1+3z^{-1}+2z^{-2}}$ for $Z > 2$				BT3/CO4	6
C)	State and prove any three properties of ZT.				BT2/CO4	6

Q. 6	Solve Any Two of the following.		12
A)	State and prove any three properties of auto-correlation.	BT2/CO5	6
B)	Compute cross correlation between $x_1(n)$ & $x_2(n)$ $X_1(n)=\{1,1,2,2,3,3\}$ & $X_2(n)=\{1,1,2,2\}$	BT4/CO5	6
C)	Write note on Energy Spectral Density & Power Spectral Density	BT2/CO5	6
	*** End ***		